

Gregory Bian

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EDUCATION

University of British Columbia

Bachelor of Applied Science, Computer Engineering

Expected May 2029

Vancouver, BC

Relevant Courses: Circuit Analysis, Computing Systems, Software Construction, Data Structures & Algorithms

TECHNICAL SKILLS

Programming & Frameworks: C, C++, Python, Java, JavaScript

Systems & Tools: Git, Docker, Linux (Ubuntu), Pytest, JUnit

Frameworks and Libraries: React, Flask, NumPy, Pandas, Matplotlib, OpenCV, BeautifulSoup/Selenium

TECHNICAL EXPERIENCE

Firmware Developer

Sep 2025 – Present

UBC Solar

Vancouver, BC

- Collaborating with a cross-disciplinary team of 80+ software, electrical, and mechanical engineers, writing firmware in C for a solar-powered car, Cascadia, in preparation for FSGP and ASC solar racing competitions.
- Coordinated with the BMS and Power & Signals subteams to calibrate the accelerator pedal across a CAN Bus to the motor, implementing fail-safes, and **ensuring precise and reliable control in a safety-critical system**.
- Led the design and implementation of a multi-page LCD Screen over a STM32 microcontroller, consulting previous drivers for layout feedback and **improving visibility of real-time driving and diagnostic data by 30%**.
- Implemented memory-robust, FreeRTOS-based task scheduling on the LCD screen to **reduce latency by 20%** and enable concurrent operation of LCD updates with driver lights and turn signal updates.

Software Developer

Jan 2025 – Jun 2025

UVIC Aero

Victoria, BC

- Configured a Linux-based flight simulator to test autonomous flight functions for Firefly, a fire-extinguishing drone; documented WSL and terminal commands to **enable knowledge transfer to future team members**.
- Engineered a Python-based hotspot detection algorithm for infrared images using thresholding and connected components analysis in OpenCV, identifying over 85% of high-intensity regions in test datasets of 30+ images.

STEM and Computing/Robotics Instructor

Jun 2025 – Aug 2025

UBC Geering Up Outreach

Vancouver, BC

- Led STEM-based Summer Camps for Geering Up, a STEM Outreach Program at the Faculty of Applied Science at UBC, reaching over 28000+ youth a year.
- Instructed up to 30 students per week on Arduino-based embedded programming, computer architecture, and data analysis in Python, explaining complex technical concepts in a fun and engaging way for youth.

PROJECTS

NotiFlow Course Tracker | *JavaScript (React), Flask, Python*

Nov 2025 – Dec 2025

- Led a team of 5 in developing a Course Tracking Chrome Extension; implemented a React-based UI and integrated Chrome Local Storage APIs to store data locally, improving usability and reliability.
- Leveraged Flask routing to send and receive data between the Chrome Extension and backend Python scripts and support features such as Canvas API access, BeautifulSoup webscraping and OpenAI API prompt filtering.

OpenCV Mini Sudoku Solver | *Python, OpenCV, Numpy*

Aug 2025 – Sep 2025

- Developed a Python script using OpenCV to automatically solve Mini Sudoku puzzles by implementing image thresholding, grid detection, digit recognition, and algorithmic solving, resulting in a 95% accurate solution.
- Resolved Optical Character Recognition issues (ie. digit misclassification, noise in blanks) through iterative experimentation with contour detection, thresholding, and adaptive cropping.

Hand Hygiene Compliance System | *MicroPython, JavaScript, Flask, Raspberry Pi*

Mar 2024 – Apr 2024

- Engineered a prototype embedded system with Raspberry Pi Pico in MicroPython to monitor Hospital Hand Hygiene Compliance of workers, resulting in a **30%** higher efficiency compared to alternate designs.
- Developed a full-stack user interface in HTML/CSS/JavaScript, featuring real-time compliance data visualization and an interactive reset button with two-way communication with Flask to backend systems.